N3 cyclicity rule | waveform review

Cycle

One detected trough to the next, lasting 0.25-2 seconds.

Amplitude

Six consecutive cycles, each ≥75 μV peak-to-peak.

Timing

SD of the six cycle durations / their mean ≤0.20. Shape may vary.

Bilateral pattern

Qualifying runs on at least one left and one right electrode must overlap in time.

The six examples show:

Strong passes (≤0.15), passes at ≤0.20, and near misses that pass only at ≤0.25.

Each example includes two channels, six-cycle boundaries, durations and peak-to-peak voltages.

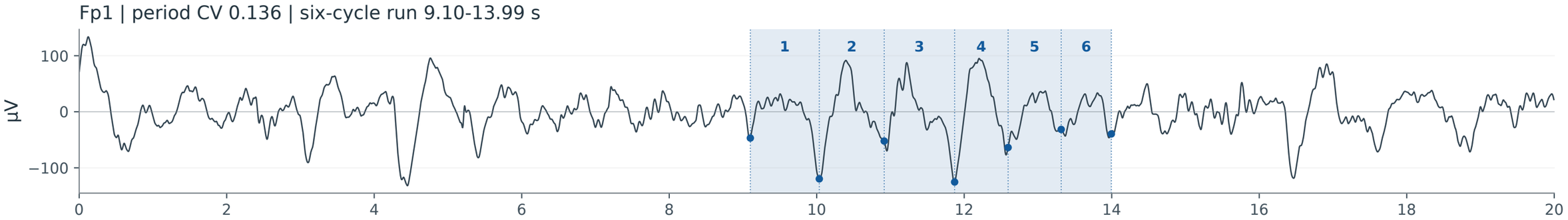
Review question

Do the highlighted waves look like sustained cyclic N3, and are the trough boundaries sensible?

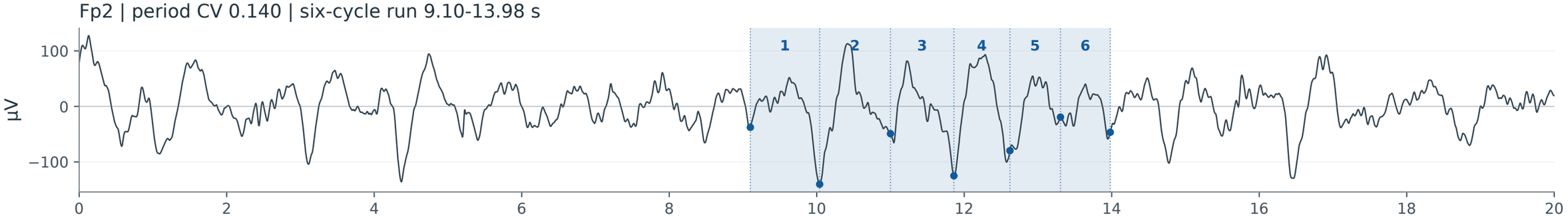
Waveform examples reviewed and accepted. The comparison uses the 0.20 timing limit.

MASS | cyclic N3 example 1

PASSES the 0.20 timing rule | Strong pass: 15%



Cycle	1	2	3	4	5	6
Period (s)	0.94	0.88	0.95	0.72	0.72	0.68
Peak-to-peak (µV)	152	211	213	220	101	80



Cycle	1	2	3	4	5	6
Period (s)	0.94	0.96	0.86	0.76	0.69	0.68
Peak-to-peak (µV)	192	253	207	219	134	99

Time from excerpt start (seconds) | EEG display: 0.5-30 Hz, common-average reference

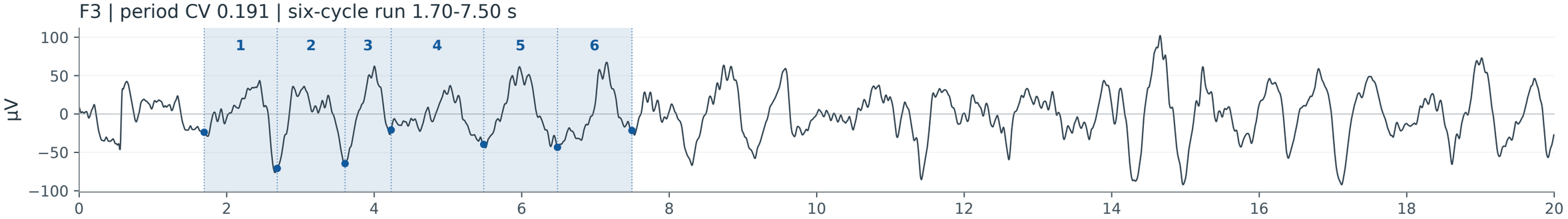
Highlighted runs overlap for 4.88 s. Total bilateral coverage at the illustrated 0.15 timing limit: 6.47 s.

Do the highlighted waves look like a sustained cyclic N3 run? Are the trough boundaries sensible?

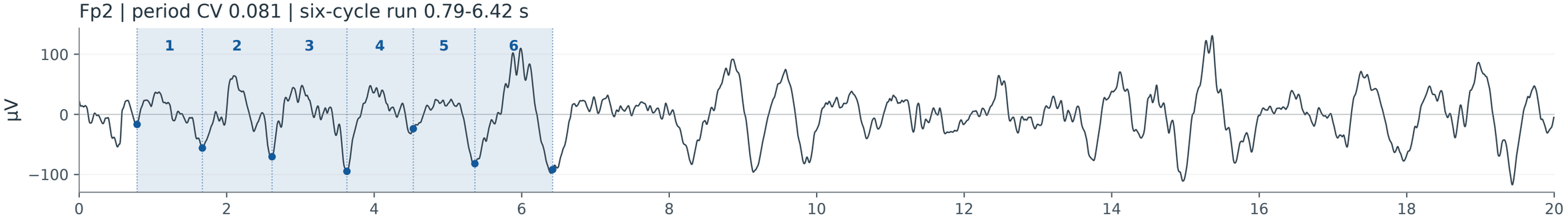
Your visual rating: Pass / Borderline / Fail

MASS | cyclic N3 example 2

PASSES the 0.20 timing rule | Passes 20%; fails 15%



Cycle	1	2	3	4	5	6
Period (s)	0.99	0.92	0.62	1.25	1.00	1.01
Peak-to-peak (µV)	120	107	127	77	106	111



Cycle	1	2	3	4	5	6
Period (s)	0.89	0.94	1.01	0.90	0.83	1.05
Peak-to-peak (µV)	93	136	143	143	111	208

Time from excerpt start (seconds) | EEG display: 0.5-30 Hz, common-average reference

Highlighted runs overlap for 4.72 s. Total bilateral coverage at the illustrated 0.20 timing limit: 4.72 s.

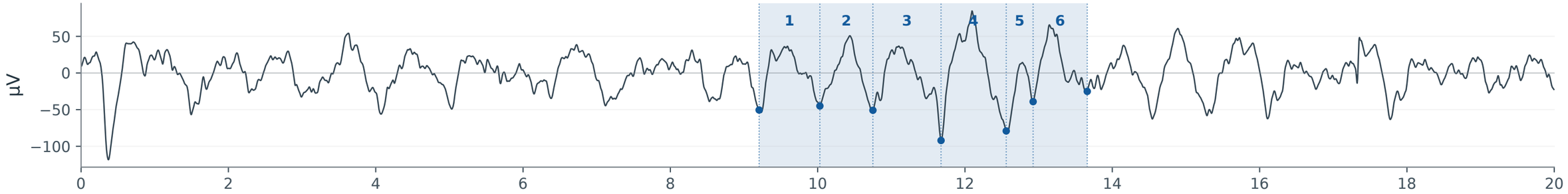
Do the highlighted waves look like a sustained cyclic N3 run? Are the trough boundaries sensible?

Your visual rating: Pass / Borderline / Fail

MASS | cyclic N3 example 3

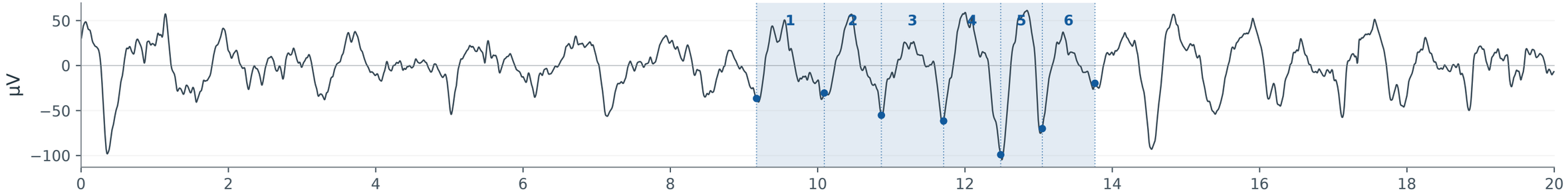
FAILS the 0.20 timing rule | Fails 20%; passes 25%

Fp1 | period CV 0.248 | six-cycle run 9.21-13.66 s



Cycle	1	2	3	4	5	6
Period (s)	0.82	0.72	0.93	0.89	0.36	0.73
Peak-to-peak (µV)	90	102	129	177	96	105

F4 | period CV 0.144 | six-cycle run 9.17-13.77 s



Cycle	1	2	3	4	5	6
Period (s)	0.92	0.78	0.84	0.78	0.56	0.71
Peak-to-peak (µV)	91	112	91	158	166	107

Time from excerpt start (seconds) | EEG display: 0.5-30 Hz, common-average reference

Highlighted runs overlap for 4.46 s. Total bilateral coverage at the illustrated 0.25 timing limit: 6.92 s.

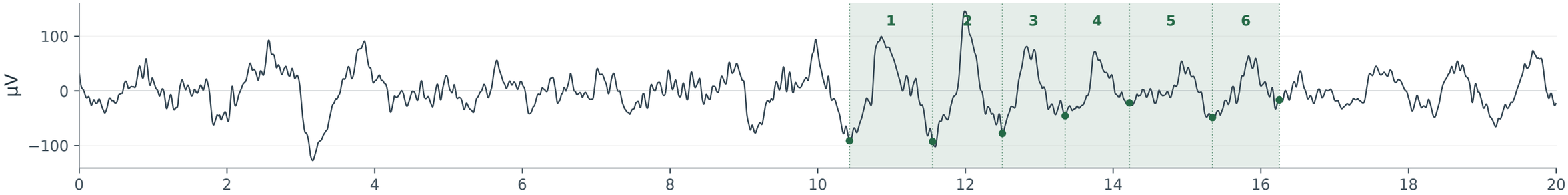
Do the highlighted waves look like a sustained cyclic N3 run? Are the trough boundaries sensible?

Your visual rating: Pass / Borderline / Fail

LENS | cyclic N3 example 1

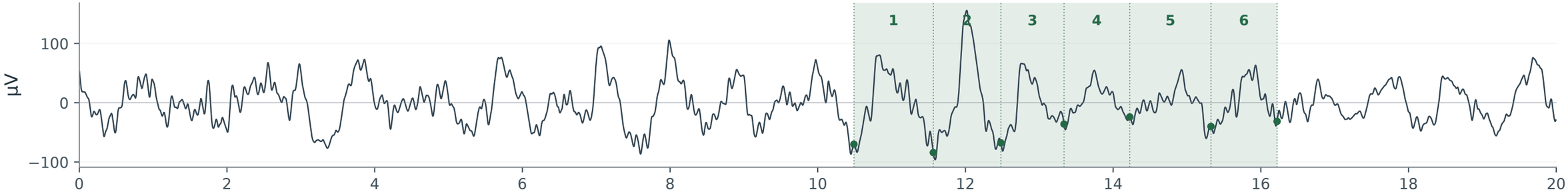
PASSES the 0.20 timing rule | Strong pass: 15%

Fp1 | period CV 0.117 | six-cycle run 10.43-16.25 s



Cycle	1	2	3	4	5	6
Period (s)	1.12	0.94	0.85	0.87	1.12	0.91
Peak-to-peak (µV)	193	249	160	117	103	117

Fp2 | period CV 0.100 | six-cycle run 10.49-16.22 s



Cycle	1	2	3	4	5	6
Period (s)	1.07	0.92	0.85	0.89	1.10	0.90
Peak-to-peak (µV)	165	252	148	100	116	115

Time from excerpt start (seconds) | EEG display: 0.5-30 Hz, common-average reference

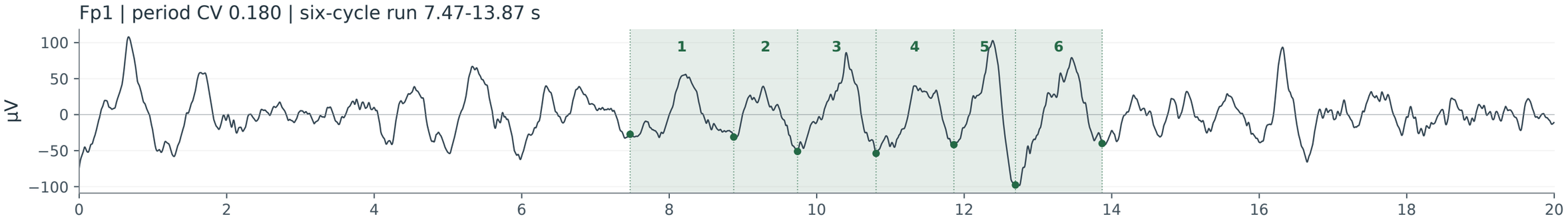
Highlighted runs overlap for 5.73 s. Total bilateral coverage at the illustrated 0.15 timing limit: 5.79 s.

Do the highlighted waves look like a sustained cyclic N3 run? Are the trough boundaries sensible?

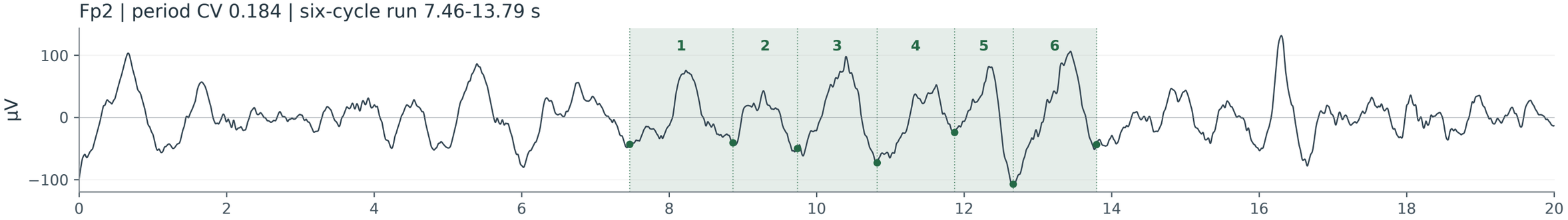
Your visual rating: Pass / Borderline / Fail

LENS | cyclic N3 example 2

PASSES the 0.20 timing rule | Passes 20%; fails 15%



Cycle	1	2	3	4	5	6
Period (s)	1.41	0.86	1.06	1.05	0.83	1.18
Peak-to-peak (µV)	88	91	140	94	202	178



Cycle	1	2	3	4	5	6
Period (s)	1.40	0.88	1.08	1.05	0.80	1.13
Peak-to-peak (µV)	119	98	172	125	189	214

Time from excerpt start (seconds) | EEG display: 0.5-30 Hz, common-average reference

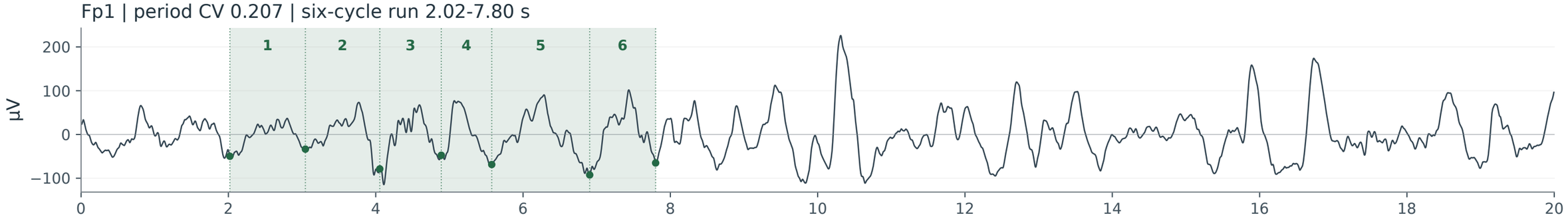
Highlighted runs overlap for 6.33 s. Total bilateral coverage at the illustrated 0.20 timing limit: 6.33 s.

Do the highlighted waves look like a sustained cyclic N3 run? Are the trough boundaries sensible?

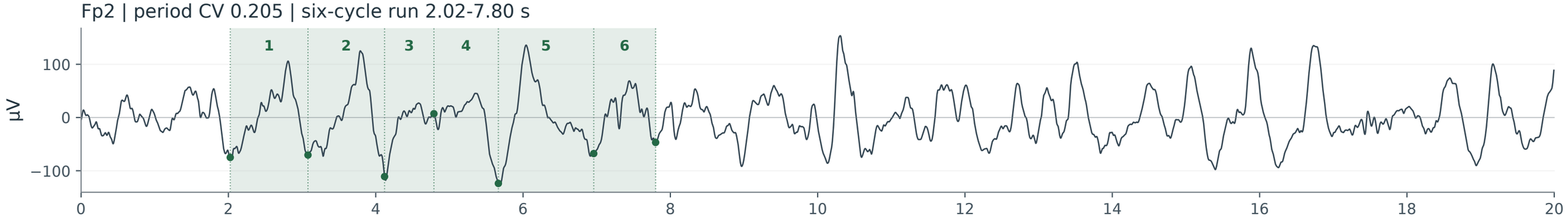
Your visual rating: Pass / Borderline / Fail

LENS | cyclic N3 example 3

FAILS the 0.20 timing rule | Fails 20%; passes 25%



Cycle	1	2	3	4	5	6
Period (s)	1.02	1.01	0.83	0.69	1.33	0.90
Peak-to-peak (µV)	89	166	182	146	184	194



Cycle	1	2	3	4	5	6
Period (s)	1.05	1.04	0.67	0.88	1.29	0.84
Peak-to-peak (µV)	181	236	145	169	262	137

Time from excerpt start (seconds) | EEG display: 0.5-30 Hz, common-average reference

Highlighted runs overlap for 5.78 s. Total bilateral coverage at the illustrated 0.25 timing limit: 7.37 s.

Do the highlighted waves look like a sustained cyclic N3 run? Are the trough boundaries sensible?

Your visual rating: Pass / Borderline / Fail